LECTURE 10b / Fall 2026

Intrinsic and Extrinsic Value of American Options

CHEME 5660
Cornell University
October 29, 2026
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LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Separate value components.** Distinguish intrinsic value, continuation value, extrinsic value, and the early-exercise premium without conflating their economic meanings.
- **Locate exercise decisions.** Use American backward induction to identify exercise and continuation regions in an option lattice.
- **Interpret market premiums.** Relate moneyness, time to expiration, volatility, rates, and dividends to observed option-value decompositions.

1 An American Option Contains a Timing Choice

An American option is not merely a terminal payoff. It is a contract that lets its holder choose an exercise time on or before expiration. That flexibility converts valuation into an optimal-stopping problem: at every admissible decision time, compare the cash flow from exercising now with the value of retaining the contract.

Let $S_{n,j} > 0$ be the underlying price at node (n, j) of a recombining lattice, let $K > 0$ be the strike in USD/share, and let $T > 0$ be expiration. The intrinsic-value functions are

$$H_c(S) := (S - K)^+, \qquad H_p(S) := (K - S)^+, \qquad x^+ := \max\{x, 0\}. \tag{1}$$

Intrinsic value is the cash flow from immediate exercise; it is not the option's full economic value before expiration.

For a step length $\Delta t > 0$, continuously compounded risk-free rate $r \in \mathbb{R}$, and risk-neutral up probability $p \in (0, 1)$, define the discounted continuation value

$$C_{n,j} := e^{-r\Delta t} \left[pV_{n+1,j+1} + (1 - p)V_{n+1,j}\right]. \tag{2}$$

Here $V_{n+1,j+1}$ and $V_{n+1,j}$ are the optimally managed American-option values at the two child nodes. The Bellman recursion is

$$V_{n,j} = \max\left\{H(S_{n,j}), C_{n,j}\right\}, \qquad V_{N,j} = H(S_{N,j}), \tag{3}$$

where H is the appropriate function from Eq. 1. Exercise is strictly optimal when $H > C$; continuation is strictly optimal when $C > H$. Equality is an indifference node, so code should state its tie-breaking convention.

Theorem 1.1. Value decomposition at an American-option node

Define extrinsic value at node (n, j) by

$$X_{n,j} := V_{n,j} - H(S_{n,j}). \tag{4}$$

Then the Bellman recursion Eq. 3 implies

$$X_{n,j} = \left(C_{n,j} - H(S_{n,j})\right)^{+} \geq 0. \tag{5}$$

Thus $X = 0$ throughout the strict exercise region and at expiration, while $X = C - H > 0$ in the strict continuation region.

Remark 1.1. Three quantities that sound similar

Continuation value C is the value of waiting. Extrinsic value $X = V - H$ is the amount by which the optimally managed contract exceeds immediate exercise value. The early-exercise premium $A := V^{\mathrm{AM}} - V^{\mathrm{EU}}$ is the value of American exercise flexibility relative to an otherwise identical European contract. In general, C, X, and A are different quantities.

2 A Two-Step American Put

Consider the textbook put example with $S_0 = 50$ USD/share, $K = 52$ USD/share, two one-year steps, $u = 1.2$, $d = 0.8$, $p = 0.6282$, and $r = 0.05$ yr^{-1}. At the one-year down node, $S = 40$ and immediate exercise pays $H_p(40) = 12$. Waiting is worth

$$C_{1,0} = e^{-0.05}\left[0.6282(4) + 0.3718(20)\right] \approx 9.46, \tag{6}$$

so exercise is optimal and $V_{1,0} = 12$. At the one-year up node, $S = 60$, intrinsic value is zero, and continuation is

$$C_{1,1} = e^{-0.05}\left[0.6282(0) + 0.3718(4)\right] \approx 1.41, \tag{7}$$

so the holder waits. Backward induction gives an American root value of approximately 5.09 USD/share, versus 4.19 USD/share for the European put. At the root, intrinsic value is 2, extrinsic value is about 3.09, and the early-exercise premium is about 0.90—a numerical demonstration of the distinctions in Remark 1.1.

COMPANION NOTEBOOK

L10b: Hull American Put Example →

Build the two-step price lattice, reproduce the American put premium, and inspect the hold-versus-exercise decision node by node.

3 Moneyness Is Not the Same as Value

Moneyness describes the relation between spot and strike. A call is in the money (ITM) when $S > K$, at the money (ATM) when $S \approx K$, and out of the money (OTM) when $S < K$; the inequalities reverse for a put. Moneyness determines intrinsic value through Eq. 1, but it does not determine the entire premium.

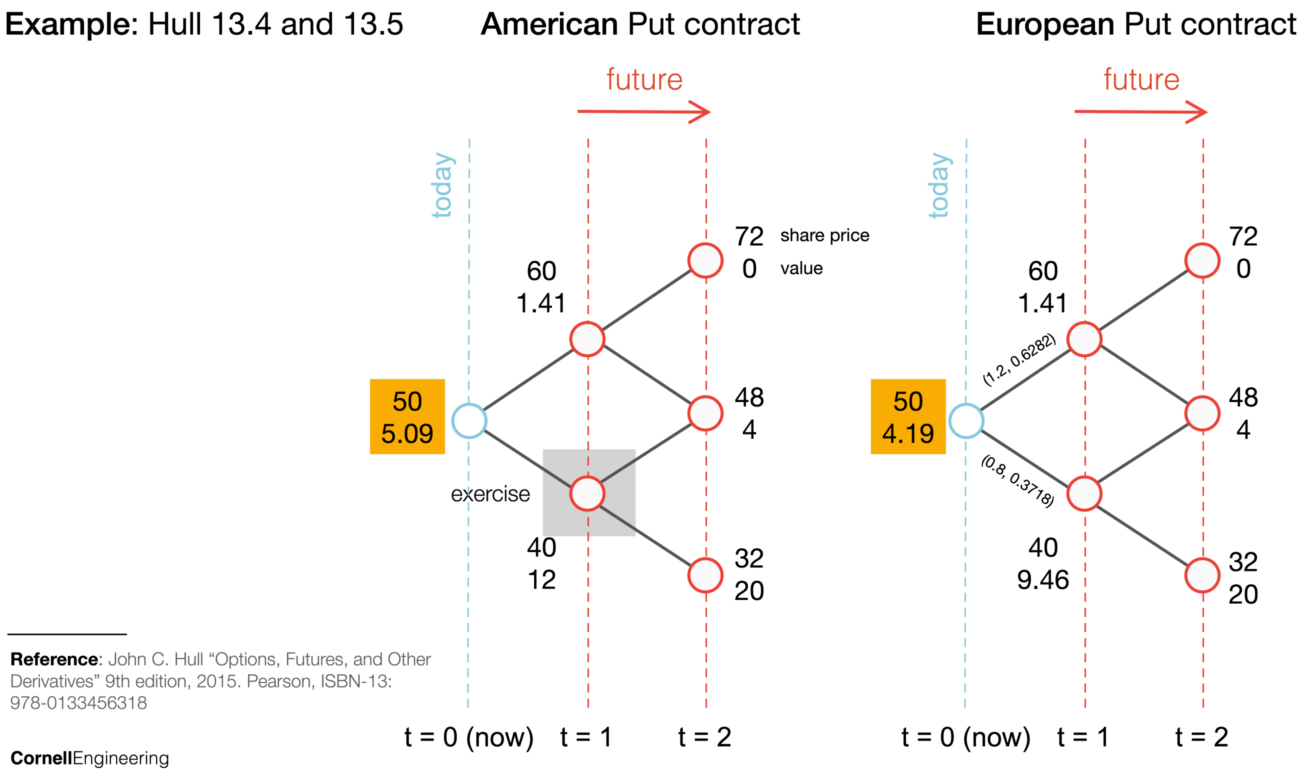

Figure 1: American and European put lattices for the two-step example. The American holder exercises at the $S = 40$ node because $12 > 9.46$; the European holder must continue. Values are in USD/share.

An OTM option has $H = 0$ yet can have $V > 0$ because future states with positive payoff remain possible. In that case the entire model value is extrinsic: $X = V$. For an ITM option, $H > 0$, but waiting can still be valuable because it preserves favorable upside states, avoids paying the strike early for a call, or delays surrendering the underlying for a put.

For fixed model inputs, extrinsic value is often largest near the exercise boundary or near the money. Far OTM options have little probability-weighted payoff; deep ITM options increasingly resemble immediate exercise or a linear position. These are qualitative tendencies, not universal monotonic laws.

> **COMPANION NOTEBOOK**
> **L10b: NVDA Intrinsic and Extrinsic Value →**
> Decompose modeled call premiums across strike and time to expiration, and connect the premium–intrinsic gap to moneyness.

4 What Moves Extrinsic Value?

Extrinsic value reflects the option's remaining opportunity set under the pricing model. Holding other inputs fixed, more time and higher volatility often increase a vanilla option's opportunity to finish favorably. Interest rates affect the value of deferring the strike payment or receipt, while dividends change the economics of owning the stock rather than the option. These effects interact with contract type and the endogenous exercise boundary.

The phrase *time decay* must be used carefully. Along a fixed hypothetical state with all other inputs held

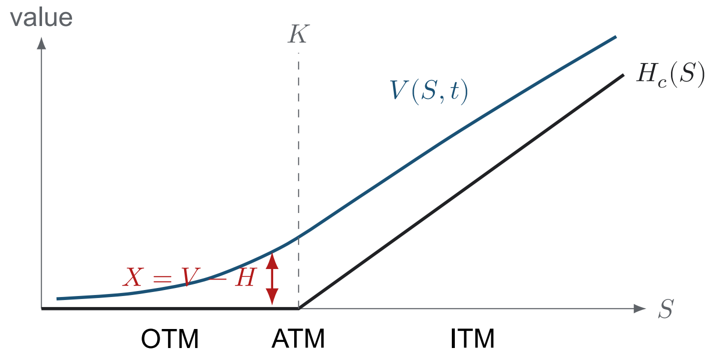

Figure 2: Schematic call-value decomposition before expiration. The blue premium curve lies above intrinsic value in an arbitrage-free American model; their vertical gap is extrinsic value. The exact shape depends on time, volatility, rates, dividends, and exercise policy.

constant, less remaining time generally removes optionality. Along an observed market path, however, S, implied volatility, rates, dividends, liquidity, and the exercise boundary all move. Consequently, a sequence of quoted extrinsic values need not decline smoothly. Theta, introduced in L11a, is a local model derivative; it is not a promise that tomorrow's market premium will be lower.

For a non-dividend-paying stock and $r \geq 0$, early exercise of an American call is suboptimal in the classical frictionless model, so $V_c^{\text{AM}} = V_c^{\text{EU}}$. American puts can optimally exercise early because receiving K sooner has financing value. Dividends can create early-exercise incentives for calls, and negative rates or trading constraints can change the standard conclusions.

5 From Model Values to an Option Chain

For a quoted market premium P^{mkt}, a descriptive decomposition uses

$$X^{\text{mkt}} := P^{\text{mkt}} - H(S). \tag{8}$$

The choice of bid, ask, last trade, or midpoint matters. Wide spreads, stale prints, asynchronous spot prices, discrete dividends, and contract multipliers can distort comparisons. A negative value from Eq. 8 is usually a data or executability warning, not evidence of negative model extrinsic value.

Likewise, a CRR price outside a quoted bid–ask spread does not by itself prove a tradable mispricing. First reconcile timestamp, spot, exercise convention, dividend assumptions, rate curve, implied volatility, tree resolution, and transaction costs. Model disagreement is a diagnostic signal; arbitrage is a stronger claim requiring an executable strategy and consistent inputs.

COMPANION NOTEBOOK
L10b: American Contract Premiums versus AMD Quotes →
Price American calls and puts across strikes with CRR and compare the model values with observed bid–ask intervals.

KEY TAKEAWAYS

In this lecture, we separated the value of exercising, waiting, and owning American exercise flexibility, then connected those quantities to lattice decisions and market quotes.

- **Extrinsic value is a residual.** $X = V - H = (C - H)^{+}$ in the American lattice; continuation value, extrinsic value, and the early-exercise premium are not synonyms.
- **Exercise is an endogenous boundary.** Backward induction chooses H or C at every node, producing exercise and continuation regions that depend on contract and market inputs.
- **Quoted decompositions require care.** Bid, ask, timestamp, dividends, volatility, and liquidity must be aligned before treating model–market differences as economic opportunities.

Next, measure how an option premium responds locally to spot, volatility, time, and interest rates through the Greeks.

ADDITIONAL READING

- John C. Cox, Stephen A. Ross, and Mark Rubinstein, "Option Pricing: A Simplified Approach," *Journal of Financial Economics* 7(3), 229–263 (1979).
- Robert C. Merton, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science* 4(1), 141–183 (1973).
- John C. Hull, *Options, Futures, and Other Derivatives*, 11th ed., Pearson (2021), chapters on binomial trees and American options.